

UNIVERSIDADE FEDERAL DO PARANÁ (UFPR)
DEPARTAMENTO DE INFORMÁTICA (DInf)
IPASP-PR PROJECT

Scholarship Holder Quarterly Activities Report

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1 Scholarship Holder Identification and Research Line

Student Name: Renan Akira N. G. Escribano
Academic Level: ☐ Undergrad (PIBIC) ☐ Undergrad (BTNM) ☐ MSc Student
☒ PhD Student ☐ Postdoc
Advisor: Luiz Eduardo Soares de Oliveira
Primary Subarea: ☐ Module 1: License Plate Detection (LPR Detection)
☐ Module 2: License Plate Legibility Classification (LPLC) and License Plate Super-Resolution (LPSR)
☐ Module 3: Character Recognition (LPR Recognition)
☐ Module 4: Fine-Grained Vehicle Classification (FGVC)
☒ Module 5: Systems Integration and Infrastructure
☐ Module 6: Facial / Ocular Recognition

2 Executive Summary of Activities

This quarter addressed the *hierarchy problem* in fine-grained vehicle classification on the project dataset UFPR-VeSV: a prediction is only usable for retrieval when **type**, **make** and **model** are *simultaneously* correct, which independent per-task heads do not guarantee. Two solutions were consolidated: Hierarchical Constrained Decoding (HCD), a training-free step that restricts the decision to the 225 tuples observed in the project ontology, and joint-class prompting, which replaces the three marginal prompt banks of a vision-language model (VLM) by a single 225-way prompt.

On eight supervised backbones, HCD raised simultaneous accuracy by +2.86 to +10.10 percentage points (pp) without breaking an already-correct triplet. On 16 zero-shot VLMs over the same five folds, the joint prompt beat paired HCD in only three backbones and lost in the other thirteen, so the decoder must be chosen per backbone. Supervising the joint head changes the regime: visual LoRA reached 94.78% (PE-Core-bigG) and 93.93% (SigLIP2-gopt), at 51k and 300k images/h on one RTX A5000. Confidence was *measured*, not assumed: calibrated HCD lowered the expected calibration error (ECE) on the six stronger supervised backbones, but on the zero-shot VLMs it improved test negative log-likelihood (NLL) in all 16 while improving ECE in only 8. Temperature scaling brought the tuned LoRA checkpoints to 0.85–1.66% ECE, the best accuracy/calibration combination obtained so far.

3 Technical Progress and Methodology

3.1 Methodology and Implementation

Task, data and metric. UFPR-VeSV contains 24,945 images of 16,297 vehicles/plates (19,573 visible, 5,372 infrared, IR) labelled with 14 **type**, 26 **make** and 136 **model** classes. Of the 49,504 Cartesian combinations only **225 tuples** occur, making the label space a two-level hierarchy [4, 5] under the surveillance conditions also reported for vehicle colour in [9]. The metric throughout is **simultaneous accuracy** (all three attributes correct on the same image), *not* the average of the three marginal accuracies. We use the five canonical folds 0..4 once each, with train/val/test separated by plate; the tables report absolute results (mean $\pm$ standard deviation over the folds), without significance testing at this stage.

HCD: constrained decoding of marginals. Given the three marginal logit vectors for an image x , raw HCD scores every admissible tuple $\tau \in \mathcal{T}$, $|\mathcal{T}| = 225$, and keeps the best one:

$$s(\tau | x) = z_{\tau_{\text{type}}}^{\text{type}} + z_{\tau_{\text{make}}}^{\text{make}} + z_{\tau_{\text{model}}}^{\text{model}}, \quad \hat{\tau} = \arg \max_{\tau \in \mathcal{T}} s(\tau | x). \quad (1)$$

This applies the label-relation constraint of [4] as pure post-processing — no retraining and no extra visual forward pass, unlike consistency-aware training [5]. The calibrated variant **HCD-C** fits two relative task weights and one joint temperature on the *validation* split of each fold by NLL; the temperature alone cannot change the **argmax**, but the weights can.

Joint 225-class prompting. For each valid tuple we build five textual descriptions, average their text embeddings and compare the image against the resulting 225 class vectors. Sixteen public checkpoints were evaluated zero-shot through OpenCLIP [2]: CLIP [10], SigLIP 1/2 [14, 11], PE-Core [1] and MetaCLIP 2 [3]. Each image is encoded **once** and the same visual embedding feeds both the marginal banks (then HCD) and the joint bank, so the two decisions are paired by checkpoint, image, pre-processing, fold and ontology; the usual “3 $\times$ cheaper” argument for joint classification therefore does *not* apply here.

Supervised joint head and calibration protocol. Four backbones were also adapted to the 225 classes on the same folds (ten epochs, training on **train**, evaluation on **test**): CoOp [15] with 16 shared context tokens and frozen visual tower, and visual LoRA [7] with rank 16, $\alpha = 32$, Q/V adaptation, batch 64 and frozen text tower. ECE is measured on the top-1 confidence of the *complete tuple* over 15 equal-width bins [8], and temperature scaling (TS) [6] and HCD-C were fitted *only* on validation. Since both objectives are NLL and neither minimises ECE directly [12], ECE is reported as measured, with NLL as the complementary criterion.

3.2 Experimental Results and Discussion

HCD on supervised backbones. In Table 1 the gain grows with the amount of hierarchically inconsistent output: 6.65% of the EfficientNet predictions and 33.00% of the YOLO11n predictions violate the ontology, and HCD rescued 750 and 2,539 cases with **zero correct triplets broken**. For an EfficientNet trained multi-task with hierarchical regularisation the inconsistency rate is already 1.26% and HCD adds only +0.48 pp (89.10 $\rightarrow$ 89.58%), bounding the benefit expected from consistency-aware training.

Zero-shot: joint prompt versus HCD. Table 2 reports the five best of the 16 VLMs. The marginal $\rightarrow$ HCD jump is very large (19.67 $\rightarrow$ 77.32% on SigLIP2-gopt): the marginal banks of a VLM are individually usable but jointly incoherent. The joint prompt wins only on

Tabela 1: HCD as post-processing on supervised single-task backbones (simultaneous accuracy). ResNet-50 (+2.97 pp) and YOLO11s (+9.70 pp) omitted for space.

Backbone	Marginal	+ HCD	Gain (pp)
EfficientNet	86.25%	89.25%	+2.86
Swin-T	86.35%	89.27%	+2.91
ResNet-101	85.55%	88.77%	+3.17
ViT	83.62%	87.09%	+3.45
MobileNet	74.58%	80.25%	+5.67
YOLO11n	44.35%	54.53%	+10.10

SigLIP2-gopt (+1.35 pp), on PE-Core-L (+0.21 pp) and, outside the top five, on MetaCLIP2-bigG (+1.61 pp); on the other 13 backbones HCD wins, by up to -13.68 pp. On the IR subgroup (5,372 images, 4,617 plates) the advantage vanishes: SigLIP2-gopt scores 69.43% either way, while PE-Core-bigG (-1.22 pp) and SigLIP2-L (-4.05 pp) are better with HCD, so the global advantage of the joint prompt must not be extrapolated to IR.

Tabela 2: Zero-shot comparison paired by image, five best of 16 VLMs (simultaneous accuracy, %). HCD uses equal weights. Δ = direct – HCD, in pp.

Backbone	Marginal	+ HCD	Direct 225	Δ (pp)
SigLIP2-gopt/16	19.67	77.32	78.66 $\pm$ 0.30	+1.35
PE-Core-bigG/14-448	51.43	75.70	74.42 $\pm$ 0.55	-1.28
SigLIP2-L/16	55.96	73.55	67.05 $\pm$ 0.47	-6.49
SigLIP2-SO400M/16	38.37	73.01	69.99 $\pm$ 0.54	-3.02
PE-Core-L/14-336	54.87	71.74	71.96 $\pm$ 0.61	$+0.21$

Supervising the joint head, and the cost frontier. Training the joint classifier changes the scale of the results (Table 3): LoRA is the best regime tested on all four backbones, +15.27 to +24.28 pp over the zero-shot joint classifier of the *same* backbone. The best zero-shot backbone is not the one that responds best to adaptation: PE-Core-bigG overtakes SigLIP2-gopt after tuning. Throughput sets the practical frontier: SigLIP2-SO400M-LoRA trades about 10% of the measured end-to-end throughput of SigLIP2-L-LoRA for +1.37 pp, whereas PE-Core-bigG-LoRA buys the last +0.85 pp over SigLIP2-gopt-LoRA at roughly $5.9\times$ the GPU time per image at batch 8 (854 versus 5,005 images/minute).

Tabela 3: Joint 225-class classifier with CoOp and LoRA (simultaneous accuracy, %; Δ in pp against the zero-shot joint classifier of the same backbone). Throughput: LoRA checkpoint, one RTX A5000, FP16, batch 8, fold 0; “GPU” excludes JPEG decoding, “JPEG $\rightarrow$ score” is the serial pipeline over 400 real JPEGs; $\sim$ marks projections.

Backbone	Zero-shot	CoOp (Δ)		LoRA (Δ)	LoRA, images/h	
	direct				GPU	JPEG $\rightarrow$ score
PE-Core-bigG	74.42	91.41 $\pm$ 0.93 (+16.99)	94.78 $\pm$ 0.48 (+20.36)	51,240	$\sim$ 43,620	
SigLIP2-gopt	78.66	79.92 $\pm$ 0.68 (+1.25)	93.93 $\pm$ 0.68 (+15.27)	300,300	$\sim$ 170,040	
SigLIP2-SO400M	69.99	71.93 $\pm$ 0.90 (+1.95)	92.71 $\pm$ 0.81 (+22.72)	702,564	242,364	
SigLIP2-L	67.05	74.54 $\pm$ 0.29 (+7.48)	91.33 $\pm$ 0.95 (+24.28)	976,890	267,720	

3.3 Confidence Calibration

Supervised reference. On the historical suite, HCD-C reduced test ECE on the six stronger backbones (EfficientNet 6.02 $\rightarrow$ 1.08%, Swin-T 6.10 $\rightarrow$ 1.07%, ResNet-101 5.94 $\rightarrow$ 0.84%, ViT 6.99 $\rightarrow$ 0.90%) but not on the two detectors (YOLO11n 3.97 $\rightarrow$ 5.98%), where raw HCD was already the best-calibrated variant even though NLL/Brier still favour HCD-C. The joint NLL ladder is consistent: EfficientNet 0.7740 (raw marginal product) $\rightarrow$ 0.5843 (raw HCD) $\rightarrow$ **0.4150** (HCD-C), and Swin-T 0.7768 $\rightarrow$ 0.6078 $\rightarrow$ **0.3969**.

Zero-shot VLMs. Closing the 80 missing validation runs (16 models $\times$ 5 folds) gives a different picture (Table 4). Direct+TS is better calibrated than HCD-C in all 16 backbones and improves on the raw direct classifier in 13 of them; the three exceptions (SigLIP2-L, SigLIP1-L, CLIP-bigG) are precisely the backbones whose raw joint confidence was already closest to calibrated, and TS was fitted for NLL, not for ECE. HCD-C reduced ECE against raw HCD in only **8 of 16** backbones while reducing test NLL in **all 16** (SigLIP2-gopt: NLL 1.31 $\rightarrow$ 1.06 with ECE 5.96 $\rightarrow$ 9.47%; SigLIP2-SO400M: NLL 1.47 $\rightarrow$ 1.21 with ECE 5.79 $\rightarrow$ 9.22%). Reweighting the tasks also moves the **argmax**, so global top-1 calibration does not follow the multiclass NLL objective. This does not establish a general superiority of the direct classifier, since its accuracy differs from that of HCD-C and tail/IR confidence was not re-evaluated here.

Tabela 4: Joint test ECE (% , 15 bins, mean over the five folds), five best zero-shot VLMs. “HCD+TS” applies one temperature to the joint score without changing the decision; HCD-C also fits two task weights and can change it (distinct fits).

Backbone	Direct raw	Direct+TS	HCD raw	HCD+TS	HCD-C
SigLIP2-gopt/16	6.15	3.56	5.96	11.55	9.47
PE-Core-bigG/14-448	14.86	2.34	4.38	10.98	9.41
SigLIP2-L/16	2.71	2.77	6.49	9.67	7.52
SigLIP2-SO400M/16	9.97	3.97	5.79	13.97	9.22
PE-Core-L/14-336	14.98	2.77	4.33	14.16	11.61

Tuned joint classifiers. The LoRA checkpoints are the only configuration that is simultaneously the most accurate and the best calibrated (Table 5): raw ECE of 1.02–2.54%, falling to 0.85–1.66% with a single temperature. CoOp instead *degrades* raw calibration with respect to its own zero-shot starting point (SigLIP2-gopt 6.15 $\rightarrow$ 17.30%, SigLIP2-SO400M 9.97 $\rightarrow$ 16.38%), which TS largely repairs. HCD-C does not apply to these checkpoints, since CoOp/LoRA produce 225-way joint logits rather than three adjustable marginal banks.

Tabela 5: Joint test ECE (% , mean over the five folds) of the eight tuned checkpoints.

Backbone	CoOp raw	CoOp+TS	LoRA raw	LoRA+TS
PE-Core-bigG	5.36	1.58	1.02	0.85
SigLIP2-gopt	17.30	3.46	2.49	1.07
SigLIP2-SO400M	16.38	3.96	2.19	1.06
SigLIP2-L	11.68	2.53	2.54	1.66

Challenges and risks. The supervised HCD study decodes over the ≈ 222 tuples of **train+val** of each fold whereas the VLM experiments use the fixed catalogue of 225: the protocols are comparable but not identical and are never merged into one paired experiment. End-to-end throughput for SigLIP2-gopt and PE-Core-bigG is still a projection and no batch-

1 latency exists yet. Aggregate ECE is bin-dependent and does not prove calibration per modality, viewpoint or tail class.

4 Quarterly Deliverables and Products

4.1 Source Code and Repositories

- **Repository URL (GitHub):** not yet published; the code is kept in the internal working tree and will be migrated to <https://github.com/UFPR-IPASP-PR/> once the ontology and protocol are frozen. **Commit ID / Tag:** not applicable.
- **Artefacts produced this quarter:** zero-shot joint evaluator and auditable prompt bank (16 models $\times$ 5 folds); CoOp/LoRA queue for the 225 classes (4 models $\times$ 2 methods $\times$ 5 folds); calibration pipeline (TS/HCD-C over 96 zero-shot variants and 8 tuned checkpoints); inference benchmark.
- **Code Status:** [x] Prototype [x] Tested/Validated [] Integrated into Main System

4.2 Scientific Publications

No publication was submitted, accepted or published in this quarter. The results of Sections 3.2 and 3.3 are being organised into a manuscript comparing constrained decoding with joint-class prompting, with the calibration analysis as a second contribution; submission is planned for the next period. Related work from the other modules is reported in [9, 13].

5 Planning for the Next Period

1. **Conditional calibration:** ECE, NLL and reliability diagrams for the eight tuned checkpoints by modality (IR/visible), viewpoint and tail classes.
2. **Statistical validation:** paired, plate-clustered comparisons for the decoder choice and for the PE-Core-bigG/SigLIP2-gopt gap, now reported as absolute values only.
3. **Complete the cost measurements:** integral JPEG $\rightarrow$ score throughput for SigLIP2-gopt/PE-Core-bigG, batch-1 latency.
4. **Hierarchy-aware calibration of the joint head:** HCD-C-style reweighting of marginals derived from the 225-way posterior, versus vector scaling.
5. **Freeze the ontology, publish the repository** (per-fold configurations, prompts, logs) and draft the manuscript.

Curitiba, 28 of September, 2026.

Student Signature

Advisor Signature

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